AIM: To create a Library Management System using Kotlin

Program:

```
import java.util.Scanner

data class Book(  3 Usages
    val id: Int,
    val title: String,
    val author: String,
    var issued: Boolean = false
)

class Library {  1 Usage
    private val books = mutableListOf<Book>()  5 Usages

    fun addBook(book: Book) {  1 Usage
        books.add(book)
        println("Book added successfully!")
    }

    fun displayBooks() {  1 Usage
        if (books.isEmpty()) {
            println("No books available.")
            return
        }
        for (b in books) {
            println("ID: ${b.id}, Title: ${b.title}, Author: ${b.author}, Issued: ${b.issued}")
        }
    }

    fun searchBook(title: String) {  1 Usage
        val found = books.filter { it.title.equals( other = title, ignoreCase = true) }
        if (found.isEmpty()) {
            println("Book not found.")
```

```
        } else {
            for (b in found) {
                println("Found -> ID: ${b.id}, Author: ${b.author}, Issued: ${b.issued}")
            }
        }
    }

    fun issueBook(id: Int) {  1 Usage
        val book = books.find { it.id == id }
        if (book != null && !book.issued) {
            book.issued = true
            println("Book issued successfully!")
        } else {
            println("Invalid ID or already issued.")
        }
    }
}

fun main() {
    val scanner = Scanner( source = System.`in`)
    val library = Library()
    var choice: Int

    do {
        println("\n--- Library Management System ---")
        println("1. Add Book")
        println("2. Display Books")
        println("3. Search Book")
        println("4. Issue Book")
        println("5. Exit")
```

```
fun main() {
    print("Enter your choice: ")

    choice = scanner.nextInt()

    when (choice) {
        1 -> {
            print("Enter Book ID: ")
            val id = scanner.nextInt()
            scanner.nextLine()

            print("Enter Title: ")
            val title = scanner.nextLine()

            print("Enter Author: ")
            val author = scanner.nextLine()

            library.addBook(Book(id, title, author))
        }

        2 -> library.displayBooks()

        3 -> {
            scanner.nextLine()
            print("Enter title to search: ")
            val title = scanner.nextLine()
            library.searchBook(title)
        }
```

```
        4 -> {
            print("Enter Book ID to issue: ")
            val id = scanner.nextInt()
            library.issueBook(id)
        }

        5 -> println("Exiting... Thank you!")

        else -> println("Invalid choice!")
    }

} while (choice != 5)
```

OP:

```
--- Library Management System ---
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Exit
Enter your choice: 1
Enter Book ID: 43
Enter Title: Shyaamchi Aai
Enter Author: R.K.kapoor
Book added successfully!

--- Library Management System ---
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Exit
Enter your choice: 2
ID: 43, Title: Shyaamchi Aai, Author: R.K.kapoor, Issued: false
```

```
--- Library Management System ---
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Exit
Enter your choice: 3
Enter title to search: Shyaamchi Aai
Found -> ID: 43, Author: R.K.kapoor, Issued: false

--- Library Management System ---
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Exit
Enter your choice: 4
Enter Book ID to issue: 43
Book issued successfully!
```

```
--- Library Management System ---
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Exit
Enter your choice: 5
Exiting... Thank you!

Process finished with exit code 0
```